Normal approximation to binomial + continuity correction

1 RECOGNISE IT

- X is a binomial count with large n.
- Exact binomial calculation is replaced by a normal model.
- Required course checks: nπ > 5 AND n(1-π) > 5.

2 DECISION / METHOD

1. Confirm nπ > 5 and n(1-π) > 5.
2. Set mean=nπ and SD=√[nπ(1-π)].
3. Apply the correct ±0.5 continuity correction.
4. Use the normal CDF/tail.

3 FORMULA

$Y \approx N(n\pi, n\pi(1-\pi))$
$SD=\sqrt{[n\pi(1-\pi)]}$
$X \le k \rightarrow Y < k+0.5$
$X \ge k \rightarrow Y > k-0.5$
$a \le X \le b \rightarrow a-0.5 < Y < b+0.5$

4 FORMULA SHEET VS WHAT I KNOW

Gives: no normal-approximation or continuity-correction rule.
Still know: conditions, normal mean/variance, and boundary shifts.

5 EXCEL ROUTE

After calculating mean and SD:
`=NORM.DIST(k+0.5,mean,SD,TRUE)`
Right tail uses 1 minus the appropriate CDF.

6 TINY WORKED EXAMPLE

If n=100, π=.40, then mean=40 and SD=sqrt(24). For P(X<=45), use `NORM.DIST(45.5,40,SQRT(24),TRUE)`.

7 TOP EXAM TRAPS

- No continuity correction.
- Wrong direction: X>k uses k+0.5, but X>=k uses k-0.5.
- Using variance where Excel wants SD.

8 SAY ALOUD

Approximate the binomial with mean n*π, SD sqrt(n*π(1-π)), then move the integer boundary by half a unit.

9 EXIT CHECK

What continuous interval represents 12<=X<=18?

ONE-GLANCE VISUAL ROUTE

